THE FROBENIUS-INDUCED ENDOMORPHISM OF THE COTANGENT COMPLEX IN CHARACTERISTIC P IS NULL HOMOTOPIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement names the characteristic hypothesis and refers to the functoriality section; the proof is a short homotopy computation.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The Frobenius homotopy on the standard resolution. Lemma lemma-frobenius-homotopy, used at proof lines 2026-2029. Replaces the given self-map by the termwise Frobenius map up to homotopy.
- The functoriality of the cotangent complex. Section section-functoriality, used at proof lines 2020-2022. Defines the map whose vanishing up to homotopy is asserted.
- The vanishing of the derivative of a p-th power. used at proof lines 2030-2032. Shows that the termwise Frobenius acts as zero on the differentials.

Source reference. Original lemma 0G5Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20178–20178. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 08QL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11064–11064. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let p be a prime number. Let $A \to B$ be a ring homomorphism and assume that $p = 0$ in A. The map $L_{B/A} \to L_{B/A}$ of Section 08QL induced by the Frobenius maps F_A and F_B is homotopic to zero.*

Proof. Let $P_\bullet$ be the standard resolution of B over A. By Lemma 0G5Y the map $P_\bullet \to P_\bullet$ induced by F_A and F_B is homotopic to the map $F_{P_\bullet} : P_\bullet \to P_\bullet$ given by Frobenius on each term. Hence we obtain what we want as clearly $F_{P_\bullet}$ induces the zero map $\Omega_{P_n/A} \to \Omega_{P_n/A}$ (since the derivative of a pth power is zero). □

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